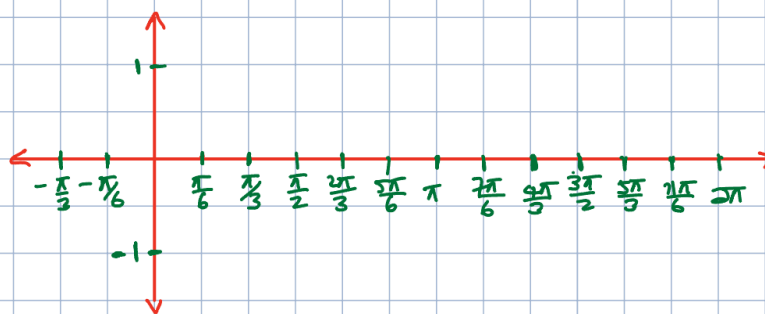


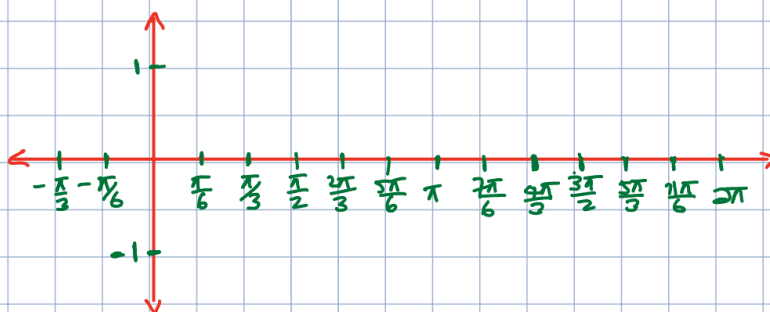
Graph of $y = \sin t$

t	$y = \sin t$	t	$y = \sin t$
$0, 2\pi$	0	π	0
$\pi/6$	$\frac{1}{2} = 0.5$	$7\pi/6$	$-\frac{1}{2} = -0.5$
$\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$	$5\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$
$\pi/3$	$\frac{\sqrt{3}}{2} = 0.9$	$4\pi/3$	$-\frac{\sqrt{3}}{2} = -0.9$
$\pi/2$	1	$3\pi/2$	-1
$2\pi/3$	$\frac{\sqrt{3}}{2} = 0.9$	$5\pi/3$	$-\frac{\sqrt{3}}{2} = -0.9$
$3\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$	$7\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$
$5\pi/6$	$\frac{1}{2} = 0.5$	$11\pi/6$	$-\frac{1}{2} = -0.5$



Graph of $y = \cos t$

t	$y = \cos t$	t	$y = \cos t$
$0, 2\pi$	1	π	-1
$\pi/6$	$\frac{\sqrt{3}}{2} = 0.9$	$7\pi/6$	$-\frac{\sqrt{3}}{2} = -0.9$
$\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$	$5\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$
$\pi/3$	$\frac{1}{2} = 0.5$	$4\pi/3$	$-\frac{1}{2} = -0.5$
$\pi/2$	0	$3\pi/2$	0
$2\pi/3$	$-\frac{1}{2} = -0.5$	$5\pi/3$	$\frac{1}{2} = 0.5$
$3\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$	$7\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$
$5\pi/6$	$-\frac{\sqrt{3}}{2} = -0.9$	$11\pi/6$	$\frac{\sqrt{3}}{2} = 0.9$



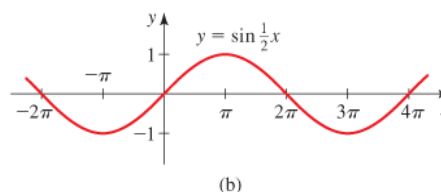
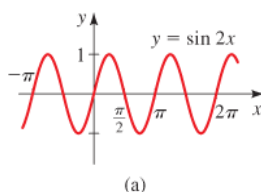
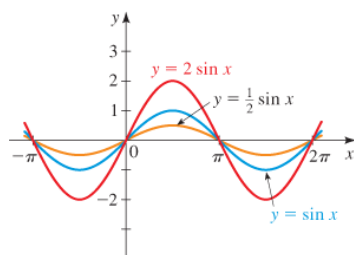
$y = 2 \sin x$
 $A = 2$
 $y = 3 \sin 4x$ ← how many complete graphs in 2π

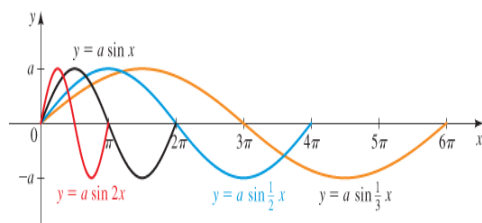
II. Graphs of Transformations of Sine and Cosine

For the functions $y = a \sin x$ and $y = a \cos x$, the number $|a|$ is called the amplitude

and is the largest value these functions attain. The sine and cosine curves $y = a \sin kx$ and $y = a \cos kx$, ($k > 0$), have amplitude $|a|$ and period $2\pi/k$. An appropriate interval on which to graph one complete period is $[0, 2\pi/k]$.

The value of k has the effect of shrinking the graph horizontally if $k > 1$ or the effect of stretching the graph horizontally if $k < 1$.





Vertical shift → Up or Down → Middle Axis
 Amplitude → UB, LB
 # of cycles or complete graphs in 2π units
 Horizontal shift, shift on x-axis
 Starting value

$$\text{Period} = \frac{2\pi}{k}$$

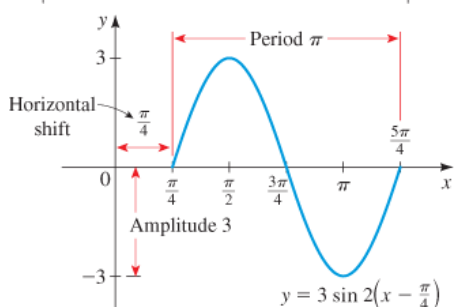
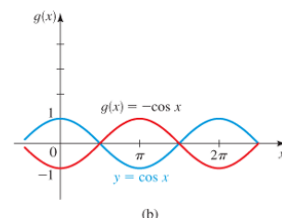
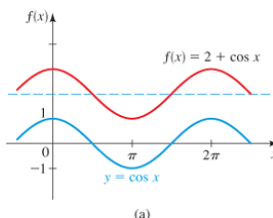
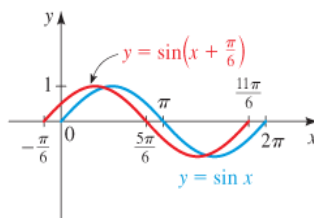
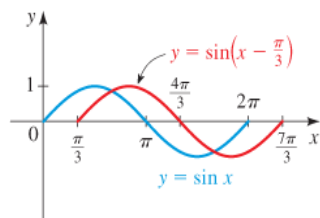
$$P = \frac{2\pi}{k}$$

$$\# \text{ of cycles is } k = \frac{2\pi}{P}$$

Shifted Sine and Cosine Curves

The sine and cosine curves $y = c + a \sin k(x - b)$ and $y = c + a \cos k(x - b)$, ($k > 0$),

have amplitude $|a|$, period $\frac{2\pi}{k}$, vertical shift c , and horizontal shift b . An appropriate interval on which to graph one complete period is $[b, b + (2\pi/k)]$.



III. Using Graphing Devices to Graph Trigonometric Functions

When using a graphing device to graph a function, it is important to choose the viewing rectangle carefully in order to produce a reasonable graph of the function.

Example 1: Graph the function, and state the domain and range. LMHML

* a. $y = 3 + \sin 2\pi(x + \frac{1}{8})$ MHMLM

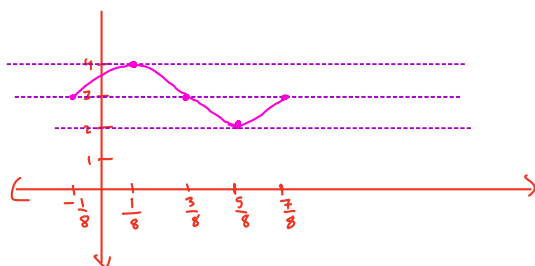
b. $y = \frac{1}{2} - \frac{1}{2} \cos(2\pi x - \frac{\pi}{3})$

1) $A = 1$

2) $c = 3$, $UB = c + A = 3 + 1 = 4$
 $LB = c - A = 3 - 1 = 2$

3) $P = \frac{2\pi}{k} = \frac{2\pi}{2\pi} = 1$

4) $b = -\frac{1}{8}$



Domain: $(-\infty, \infty)$

Range: $[2, 4]$

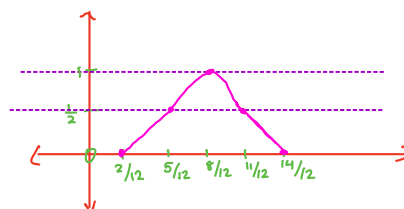
$y = \frac{1}{2} - \frac{1}{2} \cos 2\pi(x - \frac{1}{6})$

1) $A = \frac{1}{2}$

2) $c = \frac{1}{2}$, $UB = c + A = \frac{1}{2} + \frac{1}{2} = 1$
 $LB = c - A = \frac{1}{2} - \frac{1}{2} = 0$

3) $P = \frac{2\pi}{k} = \frac{2\pi}{2\pi} = 1$, $\frac{P}{4} = \frac{1}{4}$

4) $b = \frac{1}{6}$



Domain: $(-\infty, \infty)$

Range: $[0, 1]$

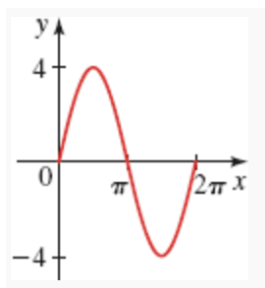
Example 2: The graph of one complete period of a sine or cosine curve is given. Find the amplitude and period, and write an equation that represents the curve in the form

$$y = a \sin k(x - b) \quad \text{or} \quad y = a \cos k(x - b)$$

1) $A = \frac{1}{2}$
 2) $C = 0$
 3) $k = \frac{2\pi}{P}$
 4) $B = -\frac{\pi}{3}$

1) $A = 4$
 2) $C = 0$
 3) $k = 1$
 4) $B = 2\pi$

a.

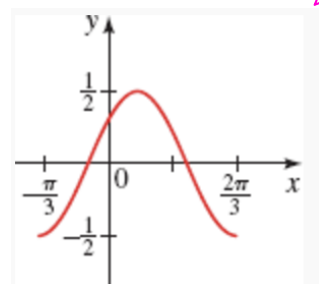


$$2\pi - 0 = 2\pi$$

$$y = 4 \sin(x)$$

$$y = -\frac{1}{2} \cos 2\left(x + \frac{\pi}{3}\right)$$

b.



$$\frac{2\pi}{3} - \left(-\frac{\pi}{3}\right) = \pi$$

Example 3: Graph the three functions on a common screen. How are the graphs related?

$$y = x^2, \quad y = -x^2, \quad y = x^2 \sin x$$

Example 4: Find the maximum and minimum values of the function.

a. $y = x - 2 \sin x, \quad (0 \leq x \leq 2\pi)$

b. $y = 2 \sin x + \sin^2 x$

Example 5: Find all solutions of the equation that lie in the interval $[0, \pi]$. State each answer rounded to two decimal places.

a. $\cos x = x \rightarrow \begin{matrix} y_1 = \cos x \\ y_2 = x \end{matrix} \text{ intersection pt}$

b. $\csc x = 3$

$$y_1 = \cos x - x$$

$$x = 0.99$$

Example 6: A tuning fork is struck, producing a pure tone as its tines vibrate. The vibrations can be modeled by the function $v(t) = 0.7 \sin(880\pi t)$ where v is the displacement of the tines in millimeters at time t seconds.

(a) Find the period of the vibration.

$$P = \frac{2\pi}{880\pi} = \frac{1}{440}$$

(b) Find the frequency of the vibration, that is, the number of times the fork vibrates per second.

$$f = \frac{1}{P} = \frac{1}{\frac{1}{440}} = 440$$

(c) Graph the function v .